Management

SWAT+ Documentation

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Purpose

The six operation databases contain named parameter sets for management operations. A scheduled operation in `management.sch` or an action in a `lum.dtl` decision table names an entry, and SWAT+ uses that entry's values when the operation runs.

File	Contents	Used by
`harv.ops`	harvest operations	**harv** and **hvkl** in `management.sch`; **harvest** and **harvest_kill** in `lum.dtl`
`graze.ops`	grazing operations	**graz** in `management.sch`; **graze** in `lum.dtl`
`irr.ops`	irrigation operations	**irrm**, **irrp** and **irpm** in `management.sch`; **irrigate** and **irr_demand** in `lum.dtl`
`chem_app.ops`	application methods for fertilizer, manure and pesticide	**fert**, **manu** and **pest** in `management.sch`; **fertilize**, **fert_future**, **manure**, **pest_apply** and **manure_demand** in `lum.dtl`
`fire.ops`	burning operations	**burn** in `management.sch` and `lum.dtl`
`sweep.ops`	street sweeping operations	**swep** in `management.sch`

The six file names are given on the **ops** line of `file.cio`, in the order harvest, graze, irrigation, chemical application, fire, sweep. The defaults are the names in the table. If a file is missing, or named **null** in `file.cio`, the database is empty, and any operation that names an entry from it uses an entry with all values zero (irrigation depth 25.4 mm).

Format

All six files share one layout. Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one operation. The columns are read by position, so every column in the tables below must be present and in this order. A trailing **description** column, which the SWAT+ Editor writes, is ignored.

Names are matched exactly against the names used in `management.sch` and `lum.dtl`. A name that is not found leaves the operation pointing at an empty entry with zero values, and no message is written.

harv.ops

#	Column	Units	Typical range	Description
1	**name**	-	-	Harvest operation name.

#	Column	Units	Typical range	Description
2	`harv_typ`	-	see below	Harvest type.
3	`harv_idx`	fraction	0 to 1	Harvest index override: the fraction of above-ground biomass cut. Used by biomass harvests and, as a multiplier, by `residue`. Ignored by grain and root harvests.
4	`harv_eff`	fraction	0 to 1	Harvest efficiency: the fraction of the harvested material removed from the HRU.
5	`harv_bm_min`	kg/ha	0 and up	Minimum plant biomass for the harvest to take place. For `residue` harvests, the minimum surface residue left behind.

The harvest types and what each one does:

`harv_typ`	Harvest only (`harv`, `harvest`)	Harvest and kill (`hvkl`, `harvest_kill`)
`grain`	grain harvest	grain harvest
`picker`	grain harvest	grain harvest
`stripper`	biomass harvest	grain harvest
`biomass`	biomass harvest	biomass harvest
`tree`	biomass harvest	no harvest; the plant is only killed
`tuber`	root harvest	root harvest
`peanuts`	root harvest	root harvest
`residue`	residue harvest	residue harvest

A harvest of a growing plant runs only if the plant's total biomass is above `harv_bm_min`. A `residue` harvest scheduled after the plant has died runs without this check.

Grain harvest. The yield is $Y = \text{harv_eff} \cdot M_{seed}$, where M_{seed} is the seed (yield) pool built up during growth from the harvest index in `plants.plt`. The seed mass that is not removed goes to the top soil layer as organic matter. `harv_idx` has no effect.

Biomass harvest. A fraction $h = \text{harv_idx} \cdot \text{harv_eff}$ of the seed, leaf and stem mass is cut. The cut takes place only if the above-ground biomass left afterwards is above `harv_bm_min`. The cut mass is reported as yield, and a further $(1 - \text{harv_eff})$ times the cut mass is added to the surface residue. Leaf area and accumulated heat units are multiplied by $1 - 0.5\,h$.

Root harvest (`tuber`, `peanuts`). The yield is `harv_eff` times the root mass. The remaining roots stay in the soil as residue.

Residue harvest. Each plant's surface residue is reduced by the fraction $e \cdot \text{harv_idx}$, but not below `harv_bm_min`. The efficiency e is the `op_data3` value of the `management.sch` row (or `const` of the decision table action) when it is greater than zero, and `harv_eff` otherwise.

For every type, the reported yield is reduced by the plant's pest stress.

graze.ops

#	Column	Units	Typical range	Description
1	`name`	-	-	Grazing operation name.
2	`fert`	-	-	Name of the manure in `fertilizer.frt` that the animals deposit. Called `fertname` in earlier documentation.
3	`bm_eat`	kg/ha/day	0 to 500	Dry biomass removed by grazing each day.

#	Column	Units	Typical range	Description
4	bm_tramp	kg/ha/day	0 to 500	Dry biomass removed by trampling each day. Read but not applied.
5	man_amt	kg/ha/day	0 to 500	Dry manure deposited each day.
6	grz_bm_min	kg/ha	0 to 5000	Minimum above-ground biomass for grazing to take place on a given day.

A graz row in management.sch starts a grazing period, and its op_data3 value is the length of the period in days. On each day of the period, if the HRU's above-ground biomass is at least grz_bm_min, the model removes bm_eat from the plants (spread over the plants in proportion to their above-ground biomass, and never more than is there) and adds man_amt of the named manure to the top soil layer. While a plant's biomass is below 2500 kg/ha, grazing also lowers its leaf area index by bm_eat/1000 and multiplies its accumulated heat units by 1 − bm_eat/1000. A graze action in a decision table applies one day of grazing each time it fires.

The parameter values in the SWAT+ Editor's default graze.ops are rough estimates. If grazing matters in a watershed, derive values from the stocking rate and type of animal. Trampling is often set equal to bm_eat (the animals trample as much as they eat), but in 62.0.1 bm_tramp has no effect.

irr.ops

#	Column	Units	Typical range	Description
1	name	-	-	Irrigation operation name.
2	amt_mm	mm	0 and up	Irrigation depth per application. Called irr_amt in earlier documentation.
3	eff_frac	fraction	0 to 1	In-field efficiency: the fraction of applied water available to the crop. Called irr_eff in earlier documentation.
4	sumq_frac	fraction	0 to 1	Fraction of the applied water that leaves as surface runoff. Called surq_rto in earlier documentation.
5	dep_sub	mm	0 and up	Application depth for subsurface irrigation. Read but not used. Called irr_dep in earlier documentation.
6	salt_ppm	ppm	0 and up	Salt concentration in the irrigation water. Used by the paddy irrigation operation irrp. Called irr_salt in earlier documentation.
7	no3_ppm	ppm	0 and up	Nitrate concentration in the irrigation water. Used by irrp. Called irr_no3n in earlier documentation.
8	po4_ppm	ppm	0 and up	Phosphate concentration in the irrigation water. Read but not used. Called irr_po4 in earlier documentation.

A scheduled irrigation (irrm or irpm in management.sch) applies

$$I_{applied} = \text{amt_mm} \cdot \text{eff_frac} \cdot (1 - \text{sumq_frac}), \qquad I_{runoff} = \text{amt_mm} \cdot \text{sumq_frac}$$

The irrigate action in lum.dtl uses the action's const (mm) in place of amt_mm and takes eff_frac and sumq_frac from the entry named in its option column. The irr_demand action with file pointer unlim uses amt_mm, but computes runoff as amt_mm · eff_frac · sumq_frac. See *.dtl for both actions.

chem_app.ops

#	Column	Units	Typical range	Description
1	name	-	-	Application method name, for example broadcast, band, foliar, inject, aerial_liquid, aerial_solid, drill, side_dress, fertigate, basal, rope_wick, tree_inject.
2	chem_form	-	solid, liquid	Form of the chemical. Read but not used.
3	app_typ	-	spread, spray, inject, direct	Application type. Read but not used.
4	app_eff	fraction	0 to 1	Application efficiency. Multiplies the amount of a pest operation or pest_apply action. Not applied to fertilizer or manure.
5	foliar_eff	fraction	0 to 1	Foliar efficiency. Read but not used.
6	inject_dp	mm	0 and up	Injection depth. Read but not used.
7	surf_frac	fraction	0 to 1	Fraction of the applied fertilizer, manure or pesticide placed in the top 10 mm soil layer. The rest goes to the second soil layer.
8	drift_pot	fraction	0 to 1	Drift potential. Read but not used.
9	aerial_unif	fraction	0 to 1	Aerial uniformity. Read but not used.

A decision table fertilize action on an HRU with standing water applies the fertilizer to the water instead of the soil when the method's surf_frac is 1.

fire.ops

#	Column	Units	Typical range	Description
1	name	-	-	Fire operation name, for example grass, tree_intense, tree_low.
2	chg_cn2	-	0 to 20	Amount added to the HRU's curve number for moisture condition II. The result is capped at 98.
3	frac_burn	fraction	0 to 1	Fraction of above-ground plant biomass, surface residue and top-layer soil organic matter burned.

A burn removes frac_burn of the above-ground biomass of every plant, of the surface residue, and of the slow and passive humus in the top soil layer. The phosphorus in the burned material is returned to the passive humus pool, and the burned plant carbon is reported as emitted carbon.

sweep.ops

#	Column	Units	Typical range	Description
1	name	-	-	Street sweeping operation name.
2	swp_eff	fraction	0 to 1	Removal efficiency: the fraction of built-up solids removed from the swept curb length.

#	Column	Units	Typical range	Description
3	frac_curb	fraction	0 to 1	Fraction of the curb length that can be swept. Parked cars and other obstacles often make part of the curb unavailable.

Street sweeping is meant for urban HRUs that use the build-up and wash-off method. The removal efficiency depends on the type of sweeper, whether flushing is part of the cleaning, the amount of solids and the frequency of rain, and is usually higher for particulate constituents than for nitrogen and phosphorus (Pitt, 1979). SWAT+ assumes fixed nutrient concentrations in the solids, so one efficiency applies to every constituent.

In 62.0.1 a `swep` operation reads the entry and logs it, but the sweeping routine uses a curb fraction of zero and is never called during the run, so `sweep.ops` has no effect on results.

Example

`refdata/Osu_1hru` in the SWAT+ repository, first rows of each file:

```
harv.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                      harv_typ      harv_idx      harv_eff   harv_bm_min  description
forest_cut                    tree       0.95000       0.99000       0.00000
grain                        grain       0.00000       0.95000       0.00000
grass_bag                  biomass       0.50000       1.00000    2000.00000
grass_mulch                biomass       0.50000       0.00000    2000.00000
```

`grass_bag` and `grass_mulch` both use a cut fraction of 0.5. With `harv_eff` at 1, `grass_bag` removes everything it cuts; with `harv_eff` at 0, `grass_mulch` cuts nothing ($h = 0.5 \times 0$). Both run only on plants above 2000 kg/ha. `grain` removes 95% of the seed pool.

```
graze.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                          fert        bm_eat      bm_tramp       man_amt    grz_bm_min  description
dairy_high                dairy_fr      25.00000      10.00000       0.95000    3000.00000  High_Prod_Dair
```

```
irr.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                       amt_mm      eff_frac     sumq_frac       dep_sub      salt_ppm       no3_ppm
sprinkler_med          25.00000       0.85000       0.00000       0.00000     100.00000      20.00000
drip                   25.00000       0.90000       0.00000     100.00000     100.00000      20.00000
```

A scheduled `sprinkler_med` irrigation puts $25 \times 0.85 = 21.25$ mm of water into the soil.

```
chem_app.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                       chem_form           app_typ       app_eff    foliar_eff     inject_dp     surf_fr
broadcast                      solid            spread       0.90000       0.00000       0.00000       1.000
inject                        liquid            inject       0.95000       0.00000     150.00000       0.200
```

```
fire.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                     chg_cn2     frac_burn  description
grass                    8.00000       1.00000
tree_low                 6.00000       0.70000
```

Related files

- `management.sch`: scheduled operations that name entries in these files.
- `*.dtl`: decision table actions that name entries in these files.
- `fertilizer.frt`: the manure named in `graze.ops`, and the fertilizers applied with a `chem_app.ops` method.
- `plants.plt`: the plant harvest index that builds the seed pool used by grain harvests.
- `file.cio`: names the six files on its `ops` line.

Sources

- Readers: mgt_read_harvops.f90, mgt_read_grazeops.f90, mgt_read_irrops.f90, mgt_read_chemapp.f90, mgt_read_fireops.f90, mgt_read_sweepops.f90, called from proc_db.f90
- Type definitions: mgt_operations_module.f90
- File name slots: `type input_ops` in input_file_module.f90
- Scheduled operations: mgt_sched.f90; decision table actions: actions.f90
- Harvest: mgt_harvgrain.f90, mgt_harvbiomass.f90, mgt_harvtuber.f90, mgt_harvresidue.f90
- Grazing: pl_graze.f90, daily loop in hru_control.f90
- Burning: pl_burnop.f90
- Street sweeping: hru_sweep.f90, hru_urbanhr.f90
- Application methods: pl_fert.f90, pest_apply.f90

File	Column	Fortran variable
harv.ops	harv_typ	harvop_db%typ
harv.ops	harv_idx	harvop_db%hi_ovr
harv.ops	harv_eff	harvop_db%eff
harv.ops	harv_bm_min	harvop_db%bm_min
graze.ops	fert	grazeop_db%fertnm (index stored in grazeop_db%manure_id)
graze.ops	bm_eat	grazeop_db%eat
graze.ops	bm_tramp	grazeop_db%tramp
graze.ops	man_amt	grazeop_db%manure
graze.ops	grz_bm_min	grazeop_db%biomin
irr.ops	eff_frac	irrop_db%eff
irr.ops	sumq_frac	irrop_db%surq
irr.ops	dep_sub	irrop_db%dep_mm
irr.ops	salt_ppm, no3_ppm, po4_ppm	irrop_db%salt, %no3, %po4
chem_app.ops	chem_form	chemapp_db%form
chem_app.ops	app_typ	chemapp_db%op_typ
chem_app.ops	inject_dp	chemapp_db%inject_dep
fire.ops	chg_cn2	fire_db%cn2_upd
fire.ops	frac_burn	fire_db%fr_burn
sweep.ops	swp_eff	sweepop_db%eff
sweep.ops	frac_curb	sweepop_db%fr_curb

Purpose

A decision table is a set of rules. Each day SWAT+ checks the table's conditions for an object (an HRU, a reservoir, a water allocation object), finds which of the table's alternatives are satisfied, and performs the actions that those alternatives switch on. Decision tables drive automatic management (planting, harvest, fertilizer, irrigation), reservoir and paddy releases, land use change scenarios, and water allocation demands. See decision tables for the concept.

There are four decision table files. Their names are given on the `decision_table` line of file.cio, in this order:

File	Default name	Used by
land use and management	lum.dtl	HRUs, through the automatic operations listed in management.sch; HRU-lte growing seasons; manure demands in manure_allo.mnu; irrigation demands, outside-source concentrations and treatment in water allocation
reservoir release	res_rel.dtl	reservoirs and HRU wetlands or paddies that name a release table (see *.res and *.wet)
land use scenarios	scen_lu.dtl	tables listed in scen_dtl.upd, checked for every HRU every day
flow control	flo_con.dtl	saturated buffers, and water allocation demand and source or receiver fractions

A file that is missing, or named `null` in file.cio, gives an empty set of tables.

Format

All four files have the same layout. Free format, whitespace-separated.

- Line 1: title (skipped).
- Line 2: the number of decision tables in the file.
- Line 3: skipped (normally blank).
- Then each table in turn:
  - a header line (skipped);
  - the table line: `name`, `conds`, `alts`, `acts` (table name, number of conditions, number of alternatives, number of actions);
  - a header line (skipped);
  - `conds` condition rows;
  - a header line (skipped);
  - `acts` action rows;
  - in `res_rel.dtl`, `scen_lu.dtl` and `flo_con.dtl` only: one more line, which is skipped. Leave a blank line here. Without it the next table's header is skipped in its place and the following lines are read out of step.

The counts on the table line must match the rows that follow. Extra or missing rows shift every later table. Names are matched exactly, and a name that is not found is not reported.

Condition rows

#	Column	Description
1	`var`	Condition variable (see Condition variables).
2	`obj`	Object type the variable belongs to, for example `hru`, `res`, `null`. For `prob_unif`, this column contains the application probability instead (see below).
3	`obj_num`	Object number. 0 means the object the table is being checked for.
4	`lim_var`	Limit variable. A name (land use, plant, soil group) or a reference volume, depending on the condition. `null` when not used.
5	`lim_op`	Limit operator: `*`, `+`, `-` or `/`. Combines `lim_var` with `lim_const` for `soil_water`, `vol` and `vol_wet`. `-` when not used.
6	`lim_const`	Limit constant.
7 and on	`alt1`, `alt2`, …	One alternative operator per alternative.

For most conditions the current value of the variable is compared with `lim_const`. For `soil_water`, `vol` and `vol_wet` it is compared with a target built from `lim_var`, `lim_op` and `lim_const`, for example "emergency volume × 0.8".

Alternatives

Each alternative is a column of operators, one per condition. An alternative is satisfied when every one of its conditions is true. For numeric conditions the operators are:

Operator	Condition is true when
`<`	value $<$ limit
`>`	value $>$ limit
`<=`	value $\leq$ limit
`>=`	value $\geq$ limit
`=`	value $=$ limit
`/=`	value $\neq$ limit
`-`	always (the condition does not apply to this alternative)

Any other text in a numeric condition's alternative column is treated like `-`. In particular `/` does not mean "not equal" for numeric conditions.

The yes/no conditions (`plant_gro`, `plant_name_gro`, `hyd_soil_group`, `land_use`, `cal_group`, `tillage`, `prob_unif`, `prob_unif1`, `prob_unif_lu`, `ch_order`) use only `=`, meaning the condition must be true. `tile_drained` and `plant_com` also accept `/`, meaning the condition must be false. Every other operator is treated like `-`.

Action rows

#	Column	Description
1	`act_typ`	Action type (see Land use and management actions and Reservoir release actions).
2	`obj`	Object type. For `irrigate`, the type of the water source.
3	`obj_num`	Object number. 0 means the object the table is being checked for.
4	`name`	Action name. Used to link `days_act` conditions to this action, and by `irrigate`, `irr_demand` and `puddle` as described below. Otherwise free text.
5	`option`	Action option, specific to the action type.
6	`const`	First action constant.
7	`const2`	Second action constant. In `lum.dtl`, values below 1 are raised to 1.
8	`fp`	File pointer: a name in another input file, specific to the action type.
9 and on	`outcome1`, `outcome2`, …	`y` or `n` for each alternative.

An action runs when at least one alternative is satisfied and the action's outcome for that alternative is `y`. Actions run in the order of the rows.

In `lum.dtl`, many actions use `const2` as the maximum number of times the action may run in a calendar year for each HRU. The count is reset on 31 December. These actions are marked "per year: `const2`" in the table below.

Condition variables

Units are those of the compared value. "Plants" means the plants of the HRU's plant community.

`var`	Value compared	Notes
`w_stress`	Average water stress factor of the growing plants (1 = no stress)	1 when no plant is growing.
`n_stress`	Average nitrogen stress factor of the growing plants	0 when no plant is growing.
`p_stress`	Average phosphorus stress factor of the growing plants	0 when no plant is growing.
`phu_plant`	Fraction of potential heat units accumulated by the first plant that has started accumulating (plant 1 if none)	
`phu_base0`	Fraction of the annual base-zero heat units accumulated at the HRU's weather station	`obj` must be `hru` or `hlt`.
`precip_cur`	Precipitation today, mm	`obj` must be `hru` or `hlt`.
`precip_next`	Precipitation tomorrow, mm	`obj` must be `hru` or `hlt`.
`plant_gro`	Plant number `lim_const` (1 if 0) is growing	Yes/no.
`plant_name_gro`	The plant named in `lim_var` is in the community and growing	Yes/no.
`days_plant`	Days since the last planting	
`days_harv`	Days since the last harvest	
`days_irr`	Days since the last irrigation	
`days_act`	Days since the action named in `lim_var` (an action `name` in the same table) last ran	Compared with `lim_const` + 2.
`day_start`	Day of year on which the simulation started	
`irr_year`	Irrigation applied to the HRU, mm	In 62.0.1 the total is not reset each year, so it is the total since the start of the run.
`slope`	HRU slope, m/m	
`soil_water`	Soil water content above wilting point, mm	Target: `lim_var` (`wp` = 0, `fc` = water at field capacity, `ul` = water at saturation), combined with `lim_const` by `lim_op`.
`jday`	Day of year	
`month`	Month	

var	Value compared	Notes
year_rot	Year of the crop rotation	Starts at the plant community's initial rotation year and increases by 1 each 31 December.
year_gro, cur_yrs_mat	Years of growth of plant 1	Same value for both names.
year_cal	Calendar year	
year_seq	Year of the simulation (1 for the first year)	
year_start	First calendar year of the simulation	
biomass	Above-ground plant biomass, kg/ha	
leaf_area	Leaf area index of all plants	
ground_cov	Above-ground plant biomass, kg/ha	Residue is not included in 62.0.1, so it gives the same value as biomass.
p_factor	USLE P factor	
k_factor	USLE K factor of the top soil layer	
hyd_soil_group	Soil hydrologic group equals lim_var (A, B, C or D)	Yes/no.
p_pet	Ratio of average annual precipitation to PET from the weather generator	
p_lab_150	Labile phosphorus in the upper soil, ppm	Uses the first two soil layers.
soil_temp2	Temperature of soil layer 2, °C	
soil_carbon	Organic carbon of soil layer 1, %	
tile_drained	The HRU has tile drains	= true, / false.
prob	A uniform random number between 0 and 1	Compared with lim_const; use < to act with probability lim_const.
prob_unif	Today is the drawn application day	See Probability conditions.
prob_unif1	Random draw within a window that starts on 1 January	See Probability conditions.
prob_unif_lu	Random draw that spreads an operation over the HRUs of a land use	See Probability conditions.
channel_flo	Flow in the current hydrograph, m^3/s	
tile_flo	Tile flow from the HRU today, mm	
irr_demand	Irrigation demand of the HRU, m^3	
irr_pou_dmd	Demand of water allocation place of use obj_num, m^3	
aqu_dep	Depth to the water table of aquifer obj_num, m	
land_use	HRU land use equals lim_var (a name in landuse.lum)	Yes/no.
cal_group	HRU calibration group in landuse.lum equals lim_var	Yes/no.
tillage	The HRU's management schedule lists the decision table named in lim_var	Yes/no.
plant_com	The plant named in lim_var is in the plant community	= true, / false.
ch_order	Stream order of channel obj_num equals lim_const	Yes/no.
wtow_vol	Water in water tower obj_num, m^3	Target: lim_const × tower capacity.
vol	Reservoir volume, m^3	Target: lim_var pvol (principal volume) or evol (emergency volume) combined with lim_const by lim_op. With + and −, lim_const is in ha · m. With lim_var e-pv, the target is $V_p + \text{lim_const}\,(V_e - V_p)$.
res_inflo	Inflow to reservoir obj_num, m^3/s	
wet_depth	Water depth of the HRU wetland or paddy	lim_const in mm.
weirh	Weir height of the HRU wetland or paddy	lim_const in mm.
vol_wet	Water volume of the HRU wetland or paddy, m^3	Target: lim_var pvol or evol combined with lim_const by lim_op.

Probability conditions

These three conditions let an operation happen on a random day within a window instead of on a fixed date.

prob_unif uses the columns differently from other conditions: obj is the probability f that the operation happens at all this year, obj_num is the first day of the window and lim_const is the last day (day of year). On the first day

of the window SWAT+ decides, with probability f, whether the operation will happen this year, and if so draws one day in the window at random. The condition is true only on that day. A window that ends before it starts (for example 330 to 30) runs over the turn of the year.

`prob_unif1` uses `obj_num` as a calendar year and `lim_const` as a number of days. The window starts on 1 January of that year. On each day of the window the condition is true with probability $1/n$, where n is the number of days left in the window.

`prob_unif_lu` uses `obj_num` as the first day and `lim_const` as the last day of the window. It is meant to be combined with a `land_use` condition in the same table: it spreads the operation over the HRUs of that land use so that the share of HRUs treated by day d follows a uniform distribution over the window.

Land use and management actions

Actions in `lum.dtl`. Unless stated otherwise, `obj` is `hru` and `obj_num` 0 means the HRU being checked.

act_typ	option	const	const2	fp	Effect
plant	plant name in the community	-	-	name in transplant.ops	Starts growth of the plant if it is not already growing. With a transplant entry, the plant starts from the transplant state (see management.sch).
harvest	plant name, or all	residue harvest efficiency	per year	name in harv.ops	Harvests the plant.
harvest_kill	plant name, or all	residue harvest efficiency	per year	name in harv.ops	Harvests and kills the plant.
kill	plant name, or all	-	per year	-	Kills the plant; biomass becomes residue.
till, tillage	name in tillage.til	-	per year	-	Tillage.
fertilize	name in fertilizer.frt	amount, kg/ha	per year	name in chem_app.ops	Applies fertilizer. On an HRU with standing water and a method with surf_frac 1, the fertilizer goes into the water.
fert_future	name in fertilizer.frt	amount, kg/ha	day of year	name in chem_app.ops	Schedules a fertilizer application on the given day. obj_num is the sequence number of the fert_future action in the table.
manure	-	amount, kg/ha	per year	-	In 62.0.1 the manure and method names are not looked up, so this action applies an empty fertilizer entry. Use fertilize with a manure from fertilizer.frt, or manure_demand.
manure_demand	name in chem_app.ops	amount, t/ha	per year	-	Sets a manure request for manure allocation. obj_num is the manure source number.
pest_apply	pesticide name listed in constituents.cs	amount, kg/ha	per year	name in chem_app.ops	Applies the amount times the method's app_eff.
graze	name in graze.ops	-	-	-	One day of grazing.
burn	name in fire.ops	-	per year	-	Burns plants and residue.

act_typ	option	const	const2	fp	Effect
irrigate	name in irr.ops	depth, mm	per year	-	Irrigates with const · eff_frac · (1 − sumq_frac). obj and obj_num name the source: aqu, cha or sdc, or res takes the water from that object and irrigates nothing if it has too little; any other obj (for example hru) is an unlimited source. With name ponding, the HRU is a paddy: const is the target ponding depth (mm), const2 the depth below which water is added, and the depth applied is the shortfall from the target.
irr_demand	name in irr.ops	depth, mm	per year	unlim or other	Sets the HRU's irrigation demand for water allocation (const× area). With fp unlim, the amt_mm of the irr.ops entry is applied directly. With name ponding, const and const2 are the target and threshold ponding depths (mm).
res_irr_dmd	-	depth, mm	-	-	Records an irrigation demand on reservoir obj_num.
rot_reset	-	rotation year	-	-	Sets the rotation year (minimum 1).
plant_reset, harv_reset, kill_reset, irr_reset	-	days	-	-	Sets the days since the last planting, harvest, kill or irrigation (minimum 1).
pheno_reset	-	-	per year	-	Restarts the phenology of plants with a moisture_gro trigger in plants.plt.
cn_update	absval, abschg, pctchg or relchg	value	per year	-	Changes the curve number for moisture condition II: set to the value, add the value, change by the value in percent, or multiply by (1 + value). The result is kept between 35 and 95.

act_typ	option	const	const2	fp	Effect
tiledep_control	-	drain depth, mm	per year	-	Sets the tile drain depth for drainage water management.
impound_on	-	-	-	name in wetland.wet	Turns on impounded water (paddy or wetland) on the HRU.
impound_off	-	-	-	-	Turns off impounded water on the HRU.
weir_height	wet, or other	height, mm	-	-	With wet, sets the HRU paddy weir height and the principal and emergency volumes from it. Otherwise sets the weir height of reservoir obj_num.
puddle	name in puddle.ops	-	per year	-	Puddles a paddy (see management.sch). The action name must be the tillage operation in tillage.til used for the soil mixing.
lu_change	-	-	-	name in landuse.lum	Changes the HRU's land use and management and reinitialises plants and curve number. Logged in lu_change_out.txt.
snow_change	-	-	-	name in snow.sno	Switches the HRU to another snow parameter set. Logged in lu_change_out.txt.
hru_fr_update	new ls_unit.ele file	-	-	new rout_unit.ele file	Updates HRU area fractions. Logged in lu_change_out.txt.
p_factor	-	new USLE P	-	-	Sets the USLE P factor.
contour	-	-	-	-	Sets $P = 2.7\,S + 0.26$, with S the slope (m/m).
stripcrop	-	-	-	pasture or other	Sets $P = 2.32\,S + 0.36$, halved when fp is pasture.
terrace	-	-	-	sod_outlet or other	Sets $P = 1.4\,S + 0.13$, doubled when fp is sod_outlet.
tile_install	-	-	-	name in tiledrain.str	Installs tile drains (see *.str).
septic_install	-	-	-	name in septic.str	Installs a septic system.
fstrip_install	-	-	-	name in filterstrip.str	Installs a filter strip.

act_typ	option	const	const2	fp	Effect
grassww_install, grassww_uninstall	-	-	-	name in grassedww.str	Installs or removes a grassed waterway.
user_def_bmp	-	-	-	name in bmpuser.str	Applies user-defined BMP removal fractions.
grow_init, grow_end	-	-	-	-	Start and end of the growing season of an HRU-lte (obj hlt). grow_end computes the yield.
chan_change	-	cover	stream order	-	Sets the cover and stream order of the channel being checked.
herd	-	-	-	-	No effect in 62.0.1.

The `plant`, `harvest` and `harvest_kill` actions of the generic tables `pl_hv_summer1`, `pl_hv_winter1` and `pl_hv_summer2` take the plant name from `management.sch` instead of `option` (see `management.sch`).

Flow control and water allocation actions, used in `flo_con.dtl`:

act_typ	option	const	const2	Effect
`tileflo_control`	`flo_mm`, `max_mm` (at most `const` mm), `min_mm` (at least `const` mm), `all_flo`, `zero_flo`, `frac` (`const` × tile flow)	mm or fraction	-	Sets the tile flow sent from a saturated buffer source HRU to its receiving HRU.
`wallo_dmd`	`set_dmd`	demand, m^3	-	Sets the demand of a water allocation place of use.
`wallo_dmd`	`irr_dmd`	-	-	Adds the irrigation demand of place of use `obj_num`.
`wallo_dmd`	`fill_res`, `fill_wtow`, `fill_can`	target fraction of capacity	fraction of the shortfall per day	Demand to fill reservoir, water tower or canal `obj_num` to `const` times its capacity (for `fill_res`, `fp` `pvol` or `evol`).
`duty_fr`	`fr1` to `fr6`	fraction	-	Fraction of the demand taken from point of diversion 1 to 6 (or sent to return 1 to 6).
`duty_reset`	-	point of diversion number	new maximum withdrawal	Resets the withdrawal limit of a point of diversion of place of use `obj_num`.

See water allocation for how these demands and fractions are used.

Reservoir release actions

In `res_rel.dtl` the action type is `release` and `option` selects how the release is computed. The releases of all actions that run on a day are added together. A table with zero alternatives runs all its actions every day. V is the current volume, V_p the principal and V_e the emergency volume.

option	const	const2	fp	Release
`rate`	rate, m^3/s	-	-	constant rate
`rate_pct`	percent	-	-	`const` % of V_p per day
`inflo_rate`	maximum rate, m^3/s	change to inflow, m^3/s	-	the larger of inflow + `const2` and `const`
`inflo_frac`	fraction	-	-	`const` × inflow
`ab_emer`	-	-	-	all volume above V_e
`days`	days	multiplier	`null`, `pvol` or `evol`	$(V - V_t)$/const, with target $V_t = 0$, `const2` × V_p or `const2` × V_e
`dyrt`	days	percent	`null`, `pvol` or `evol`	$(V - V_t)$/const plus `const2` % of V_p, with V_t = 0, V_p or V_e
`dyrt1`	days	percent	`con1`, `con2` or `con3`	as `dyrt`, with the target taken from condition 5, 4 or 3 of the table
`inflo_targ`	condition number	-	-	inflow plus all volume above V_p × that condition's `lim_const`

option	const	const2	fp	Release
weir	-	-	name in weir.res	weir equation
meas	-	-	name of a recall object	measured outflow
natlake	inactive storage, fraction of V_p	release coefficient, 1/day	-	natural lake outflow (Döll et al., 2003)
nonirr-h06	α	-	-	Hanazaki et al. (2006) scheme for non-irrigation reservoirs
irr-h06	α	β	-	Hanazaki et al. (2006) scheme for irrigation reservoirs
hydrop	amplitude	phase	-	HYPE hydropower scheme (Arheimer et al., 2019)

See *.res for the reservoir side of these options.

Land use scenarios

Tables in scen_lu.dtl run only if they are listed in scen_dtl.upd. That file has a title line, the number of entries, a header line, and one row per entry with three columns: max_hits, typ and the name of a table in scen_lu.dtl. Each listed table is checked for every HRU every day, before the HRUs are simulated. The usual actions are lu_change and snow_change.

Example

refdata/Ames_sub1/lum.dtl in the SWAT+ repository contains two tables:

```
lum.dtl: yearly irrigation limit example for Ames dataset
2

name                      conds      alts      acts       !limit yearly irrigation to 500 mm using irrigate
irr_year_irr                 4         1         1
var                        obj   obj_num           lim_var            lim_op  lim_const      alt1
irr_year                   hru         0              null                 -   500.00000         <
w_stress                   hru         0              null                 -   0.80000           <
phu_plant                  hru         0            phu_mat                -     0.20000         >
leaf_area                  hru         0              null                 -     0.50000         >
act_typ                    obj   obj_num              name            option    const        const2
irrigate                   hru         0          sprinkler     sprinkler_ilm    25.00000   20.00000  unli
```

Table irr_year_irr has four conditions, one alternative and one action. The alternative is satisfied on a day when the irrigation applied so far is below 500 mm, the average water stress factor is below 0.8, the crop has accumulated more than 20% of its heat units, and the leaf area index is above 0.5. On such a day the HRU is irrigated with 25 mm using the sprinkler_ilm entry of irr.ops, at most 20 times a year. obj is hru, so the water comes from an unlimited source. The table is used by naming irr_year_irr as an automatic operation in management.sch. In 62.0.1 the irrigation total behind irr_year is not reset each year, so once 500 mm has been applied over the whole run, the table stops irrigating for the rest of the simulation.

A reservoir table from refdata/Osu_1hru/res_rel.dtl, followed by the blank line that res_rel.dtl needs:

```
name                      conds      alts             acts
paddy_auto_rel               2         2                2
var                        obj   obj_num           lim_var            lim_op     lim_const      alt1   alt
wet_depth                  hru         0          weir_hgt                 -       0.00000         >
wet_depth                  hru         0              null                 -       2.00000         >
```

act_typ	obj	obj_num	name	option	const	const2
release	hru	0	above_emer	days	1.00000	0.00000
release	hru	0	below_emer	weir	10.00000	0.00000

Both alternatives need a water depth above 2 mm, and the first also above 0 mm. When the first alternative is satisfied, the paddy releases its whole volume within one day (`days` with `const` 1 and target 0) and also releases over the weir `weir1`.

Related files

- `management.sch`: lists the `lum.dtl` tables each HRU uses.
- `*.ops`, `tillage.til`, `fertilizer.frt`, `landuse.lum`, `snow.sno`, `*.str`: databases named by actions.
- `*.res` and `*.wet`: name the release tables in `res_rel.dtl`.
- Water allocation and manure allocation: name tables in `lum.dtl` and `flo_con.dtl`.
- `file.cio`: names the four files on its `decision_table` line.

Sources

- Readers: `dtbl_lum_read.f90`, `dtbl_res_read.f90`, `dtbl_scen_read.f90`, `dtbl_flocon_read.f90`, `cal_cond_read.f90` (`scen_dtl.upd`)
- Type definitions: `conditional_module.f90`
- Conditions: `conditions.f90`, `cond_real.f90`, `cond_integer.f90`
- Actions: `actions.f90`; reservoir releases: `res_hydro.f90`
- Linking tables to HRUs: `proc_cond.f90`, `hru_dtbl_actions_init.f90`
- Callers: `hru_control.f90`, `time_control.f90`, `res_control.f90`, `wallo_start_day.f90`, `mallo_control.f90`
- File name slots: `type input_condition` in `input_file_module.f90`

Column	Fortran variable
condition `var`, `obj`, `obj_num`, `lim_var`, `lim_op`, `lim_const`	`d_tbl%cond(ic)%var`, `%ob`, `%ob_num`, `%lim_var`, `%lim_op`, `%lim_const`
`alt1` …	`d_tbl%alt(ic, ialt)`
`prob_unif` probability (`obj` column)	`d_tbl%frac_app`
action `act_typ`	`d_tbl%act(iac)%typ`
action `obj`, `obj_num`, `name`, `option`, `const`, `const2`	`d_tbl%act(iac)%ob`, `%ob_num`, `%name`, `%option`, `%const`, `%const2`
action `fp`	`d_tbl%act(iac)%file_pointer`
`outcome1` …	`d_tbl%act_outcomes(iac, ialt)`

Purpose

`plants.plt` is the plant growth database. Each row is one plant species, crop variety or land cover, with the parameters SWAT+ needs to grow it: plant type, heat units to maturity, leaf area development, radiation use efficiency, canopy height, rooting depth, temperature limits, nutrient content, stomatal conductance and residue properties.

Plant communities in `plant.ini`, plant operations in `management.sch` and decision tables, and land uses in `landuse.lum` all refer to rows of this file by name. The names are also used by the GIS interfaces to link land cover maps to SWAT+ land uses.

The file is named on the `hru_parm_db` line of `file.cio` (default `plants.plt`). If the file is missing, the database is empty.

The database distributed with SWAT+ has parameters for most common crops, grasses, trees and land covers. If a land use or plant is not in it, start from a row for a plant with similar growth habits.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one plant with 56 values in the order below. The reader does not match columns by header name, so every value must be present and in this order. It ignores anything after the 56th value, so the `description` column written by the SWAT+ Editor has no effect.

If the `event` column of `codes.bsn` is not 0, one more word is read after the 56th value: a plant class name of up to 25 characters (for example `row_crop`). In that case the plant class takes the place of the description.

#	Column	Units	Default	Description
1	`name`	-	-	Name of the plant or land cover, up to 40 characters. Must be unique.
2	`plnt_typ`	-	-	Plant type: `warm_annual`, `cold_annual`, `warm_annual_tuber`, `cold_annual_tuber` or `perennial`. See Plant types.
3	`gro_trig`	-	-	Growth trigger: `temp_gro` (growth driven by temperature and day length) or `moisture_gro` (tropical plants whose growth starts with the monsoon).
4	`nfix_co`	-	0.0	Nitrogen fixation coefficient: 0.5 for legumes, 0 for non-legumes. Any value above 0 lets the plant fix nitrogen when soil nitrate cannot meet its demand.
5	`days_mat`	days	110	Days to maturity, an integer. Used to compute the heat units to maturity. 0 means the plant uses the heat units of the whole year. See Heat units to maturity.
6	`bm_e`	(kg/ha)/(MJ/m²)	15.0	Biomass-energy ratio (radiation use efficiency): dry biomass produced, including roots, per unit of intercepted photosynthetically active radiation, with no stress, at ambient CO2 and a vapor pressure deficit of 1 kPa.
7	`harv_idx`	(kg/ha)/(kg/ha)	0.76	Harvest index for optimal growing conditions: fraction of above-ground biomass removed as yield. Values above 0.7 are replaced by 0.02 (see Values set by the model).
8	`lai_pot`	m²/m²	5.0	Maximum potential leaf area index. Values are limited to 0 to 10.
9	`frac_hu1`	fraction	0.05	Fraction of heat units to maturity at the 1st point of the optimal leaf area development curve.
10	`lai_max1`	fraction	0.05	Fraction of `lai_pot` at the 1st point of the optimal leaf area development curve.
11	`frac_hu2`	fraction	0.40	Fraction of heat units to maturity at the 2nd point of the optimal leaf area development curve.
12	`lai_max2`	fraction	0.95	Fraction of `lai_pot` at the 2nd point of the optimal leaf area development curve.
13	`hu_lai_decl`	fraction	0.99	Fraction of heat units to maturity at which leaf area starts to decline.
14	`dlai_rate`	-	1.0	Exponent of the leaf area decline of annuals after `hu_lai_decl`. 1 gives a linear decline.
15	`can_ht_max`	m	6.0	Maximum canopy height.
16	`rt_dp_max`	m	3.5	Maximum rooting depth. The soil profile depth also limits roots.
17	`tmp_opt`	°C	30.0	Optimal temperature for plant growth.
18	`tmp_base`	°C	10.0	Base (minimum) temperature for plant growth. Heat units accumulate only above it.
19	`frac_n_yld`	kg N/kg yield	0.0015	Fraction of nitrogen in the yield, dry weight.

#	Column	Units	Default	Description
20	`frac_p_yld`	kg P/kg yield	0.0003	Fraction of phosphorus in the yield, dry weight.
21	`frac_n_em`	kg N/kg biomass	0.006	Normal fraction of nitrogen in plant biomass at emergence.
22	`frac_n_50`	kg N/kg biomass	0.002	Normal fraction of nitrogen in plant biomass at 50% maturity.
23	`frac_n_mat`	kg N/kg biomass	0.0015	Normal fraction of nitrogen in plant biomass at maturity.
24	`frac_p_em`	kg P/kg biomass	0.0007	Normal fraction of phosphorus in plant biomass at emergence.
25	`frac_p_50`	kg P/kg biomass	0.0004	Normal fraction of phosphorus in plant biomass at 50% maturity.
26	`frac_p_mat`	kg P/kg biomass	0.0003	Normal fraction of phosphorus in plant biomass at maturity.
27	`harv_idx_ws`	(kg/ha)/(kg/ha)	0.01	Harvest index under severe water stress: the lowest harvest index expected, between 0 and `harv_idx`.
28	`usle_c_min`	-	0.001	Minimum USLE C factor for water erosion. Values are limited to 0.001 to 1.
29	`stcon_max`	m/s	0.002	Maximum stomatal conductance at high solar radiation and low vapor pressure deficit. Used with the Penman-Monteith ET method.
30	`vpd`	kPa	4.0	Vapor pressure deficit at the 2nd point of the stomatal conductance curve.
31	`frac_stcon`	fraction	0.75	Fraction of `stcon_max` at the 2nd point of the stomatal conductance curve.
32	`ru_vpd`	(kg/ha)/(MJ/m²)/kPa	8.0	Rate of decline in radiation use efficiency per unit increase in vapor pressure deficit above 1 kPa. 0 turns the adjustment off.
33	`co2_hi`	µL CO2/L air	660.0	Elevated CO2 concentration at the 2nd point of the radiation use efficiency curve.
34	`bm_e_hi`	(kg/ha)/(MJ/m²)	16.0	Biomass-energy ratio at the CO2 concentration `co2_hi`.
35	`plnt_decomp`	-	0.05	Plant residue decomposition coefficient: fraction of residue that decomposes in a day under optimal moisture, temperature, C:N and C:P. 0 takes `rsd_decomp` from `parameters.bsn`.
36	`lai_min`	m²/m²	0.75	Minimum leaf area index of perennials during the dormant period.
37	`bm_tree_acc`	-	0.3	Exponent that limits the maximum leaf area index of young perennials. See Perennials and trees.
38	`yrs_mat`	years	10	Years for a perennial to reach full development, an integer. Values below 1 are set to 1.
39	`bm_tree_max`	t/ha	1000.0	Maximum above-ground biomass. Checked once a year.
40	`ext_co`	-	0.65	Light extinction coefficient for intercepted photosynthetically active radiation. 0 is replaced by 0.65.
41	`leaf_tov_mn`	months	12.0	Leaf turnover time of `moisture_gro` perennials with minimum stress. Called `leaf_tov_min` in earlier documentation.
42	`leaf_tov_mx`	months	3.0	Leaf turnover time of `moisture_gro` perennials with maximum stress. Called `leaf_tov_max` in earlier documentation.
43	`bm_dieoff`	fraction	0.0	Factor on the excess biomass removed when above-ground biomass exceeds `bm_tree_max`. 0 is replaced by 1.
44	`rt_st_beg`	fraction	0.0	Root to shoot ratio at the start of the growing season. 0 is replaced by 0.4.

#	Column	Units	Default	Description
45	rt_st_end	fraction	0.0	Root to shoot ratio at the end of the growing season. 0 is replaced by 0.2. Tuber types always use 0.7.
46	plnt_pop1	plants/m²	0.0	Plant population at the 1st point of the population and leaf area curve.
47	frac_lai1	fraction	0.0	Fraction of lai_pot at the 1st point of the population and leaf area curve.
48	plnt_pop2	plants/m²	0.0	Plant population at the 2nd point of the population and leaf area curve.
49	frac_lai2	fraction	0.0	Fraction of lai_pot at the 2nd point of the population and leaf area curve.
50	frac_sw_gro	-	0.5	Intended as the 30-day precipitation to PET ratio that starts growth of moisture_gro plants. Read but not used, which uses a fixed 0.5.
51	aeration	fraction	0.2	Aeration stress threshold: the fraction of pore space above field capacity that can fill with water before the plant is stressed by lack of air. 0 is replaced by 0.2.
52	rsd_pctcov	-	0.0	Residue factor for the percent cover equation. 0 is replaced by 0.4.
53	rsd_covfac	-	0.0	Residue factor for the surface cover (C factor) equation. 0 is replaced by 0.04.
54	avg_lig_frac	fraction	0.85	Average lignin fraction of residue. Read but not used.
55	ab_lig_frac	fraction	0.15	Lignin fraction of above-ground residue. Used by the dynamic carbon model only.
56	bg_lig_frac	fraction	0.12	Lignin fraction of below-ground residue (roots). Used by the dynamic carbon model only.

The defaults are the values SWAT+ uses for a component that is not set. They are not filled in for missing columns, since every row must have all 56 values.

Plant types

plnt_typ decides how the plant goes through the year.

- warm_annual and cold_annual are annual crops. A cold_annual is planted in the fall and goes dormant over winter when day length falls below the dormancy threshold. Leaf area declines after hu_lai_decl, following dlai_rate.
- warm_annual_tuber and cold_annual_tuber are annuals whose yield is below ground. Their root to shoot ratio at the end of the season is fixed at 0.7.
- perennial covers grasses, shrubs and trees. Perennials go dormant when day length falls below the threshold, keep at least lai_min of leaf area, and develop over yrs_mat years.

gro_trig = moisture_gro is for tropical plants. If the plant was planted with the monsoon option, it starts growing when the 30-day ratio of precipitation to potential ET reaches 0.5. Leaf turnover of moisture_gro perennials varies between leaf_tov_mx and leaf_tov_mn months with moisture stress.

Heat units to maturity

SWAT+ computes the heat units a plant needs to reach maturity at the start of the run, from the monthly mean temperatures of the HRU's weather generator station.

- days_mat = 0: the plant grows through the whole year. Heat units to maturity are 95% of the year's sum of mean daily temperature above tmp_base.
- days_mat > 0 for an annual: the planting date is taken as the day when the sum of mean daily temperatures above 0 °C reaches 15% of the annual sum. Heat units above tmp_base are then summed for days_mat days from that date.

The calculation works in both hemispheres. Giving the days of a variety (for example 90, 100, 110 or 120 days for corn) lets one database support several varieties. If `days_mat` is longer than the growing season at a site, the heat units of the whole season are used.

Leaf area development

The optimal leaf area index follows a curve through two points, (`frac_hu1`, `lai_max1`) and (`frac_hu2`, `lai_max2`), where the first value is the fraction of heat units to maturity and the second the fraction of `lai_pot`:

$$f_{LAI} = \frac{f_{PHU}}{f_{PHU} + \exp(l_1 - l_2\, f_{PHU})}$$

l_1 and l_2 are set at the start of the run so that the curve passes through both points. The values of `lai_pot` in the standard database are for typical plant densities in rainfed agriculture. Drought-prone regions with low planting densities, or irrigated land with high densities, may need different values. Canopy height grows with the square root of f_{LAI} up to `can_ht_max`.

When `plnt_pop1` and `plnt_pop2` are given, a second curve through (`plnt_pop1`, `frac_lai1`) and (`plnt_pop2`, `frac_lai2`) limits `lai_pot` by plant population.

Radiation use efficiency

When the `co2` column of `parameters.bsn` is above 350 µL/L, radiation use efficiency is adjusted for CO2 along a curve through (330 µL/L, `bm_e`) and (`co2_hi`, `bm_e_hi`). Otherwise `bm_e` is used as given. Values for `co2_hi` and `bm_e_hi` are needed even when climate change is not simulated, and `bm_e_hi` should be larger than `bm_e`. A `co2_hi` of 330 is replaced by 660.

When the vapor pressure deficit is above 1 kPa, radiation use efficiency drops by `ru_vpd` for each kPa above 1, down to no less than 27% of `bm_e`. A value of 6 to 8 is a reasonable approximation for most plants, and 0 turns the adjustment off.

Perennials and trees

The maximum leaf area index of a perennial grows with its age:

$$LAI_{max} = \text{lai_pot} \left(\frac{y}{\text{yrs_mat}} \right)^{\text{bm_tree_acc}}$$

where y is the number of years since the plant was established (at least 1), and LAI_{max} is at most `lai_pot`. With `bm_tree_acc` = 0 the plant reaches `lai_pot` in its first year.

At the end of each year, if above-ground biomass B exceeds `bm_tree_max` (converted to kg/ha), the amount

$$D = (1 + \text{bm_dieoff})\,(B - 1000\,\text{bm_tree_max})$$

dies and goes to the residue pools, with roots in the same proportion. Nothing dies if D is not smaller than B. A mature forest stand typically has 30 to 50 t/ha of biomass. `bm_dieoff` is not used at the start of dormancy in 62.0.1.

Nutrient uptake

The six fractions `frac_n_em` to `frac_p_mat` describe the nitrogen and phosphorus content of the whole plant, roots included, on a dry weight basis, at emergence, at 50% maturity and at maturity. They set the plant's nutrient demand through the season. Each fraction must be smaller than the one before it: if `frac_n_50` is not at least 0.0001 below `frac_n_em`, it is set to `frac_n_em` − 0.0001, and if `frac_n_mat` is not at least 0.0001 below `frac_n_50`, it is multiplied by 0.75. The phosphorus fractions are treated the same way.

Residue lignin

With `carbon` = 2 in codes.bsn, residue entering the soil is split into metabolic, structural and lignin pools. Below ground, the lignin fraction is `bg_lig_frac`, the structural fraction is `bg_lig_frac` / 0.8, and the rest is metabolic. Above ground the same split uses `ab_lig_frac`. With `carbon` = 0 these columns are not used.

Values set by the model

After the file is read, SWAT+ changes some values:

Column	Change
`harv_idx`	Values above 0.7 are set to 0.02.
`lai_pot`	Limited to 0 to 10.
`usle_c_min`	Limited to 0.001 to 1.
`plnt_decomp`	0 is replaced by `rsd_decomp` of parameters.bsn.
`yrs_mat`	Values below 1 are set to 1.
`ext_co`	0 is replaced by 0.65.
`bm_dieoff`	0 is replaced by 1.
`rt_st_beg`, `rt_st_end`	0 is replaced by 0.4 and 0.2. Tuber types get `rt_st_end` = 0.7.
`aeration`, `rsd_pctcov`, `rsd_covfac`	0 is replaced by 0.2, 0.4 and 0.04.
`co2_hi`	330 is replaced by 660.
`frac_n_50`, `frac_n_mat`, `frac_p_50`, `frac_p_mat`	Adjusted to decrease, as described above.

The `harv_idx` change affects crops whose harvest index is above 0.7, such as alfalfa (0.9 in the standard database) and tuber crops with a harvest index above 1. Their harvest index becomes 0.02.

Calibration

These parameters can be changed during calibration through calibration.cal: `bio_e` (`bm_e`), `hvsti` (`harv_idx`), `rdmx` (`rt_dp_max`), `t_opt` (`tmp_opt`), `t_base` (`tmp_base`) and `usle_c` (`usle_c_min`).

Example

The first rows of `refdata/Ames_sub1/plants.plt` in the SWAT+ repository:

```
plants.plt
name  plnt_typ  gro_trig  nfix_co  days_mat  bm_e  harv_idx  lai_pot  frac_hu1  lai_max1  frac_hu2  lai_ma
alfa  perennial  temp_gro  0.5  0  20  0.9  4  0.15  0.01  0.5  0.95  0.9  0.5  0.9  3  20  4  0.02  0.003
cana  warm_annual  temp_gro  0  100  34  0.3  4.5  0.15  0.02  0.45  0.95  0.5  0.3  1.3  1.4  21  5  0.03
```

- `alfa` (alfalfa) is a perennial legume (`nfix_co` = 0.5) with `days_mat` = 0, so it uses the heat units of the whole year. Its `harv_idx` of 0.9 is replaced by 0.02.
- `alfa` has `frac_n_50` = `frac_n_mat` = 0.02, so `frac_n_mat` is multiplied by 0.75 to 0.015.
- `bm_tree_max` is 0 for both rows. With `bm_dieoff` = 0.1, $D = 1.1\,B$ at the end of the year, which is not smaller than B, so no biomass dies off.
- Ames_sub1 has `event` = 0 and `carbon` = 2 in codes.bsn. No plant class is read, so the `description` column is ignored, and the lignin columns are used: for `alfa`, 9% of decomposing surface residue and 20% of root residue entering the soil goes to the lignin pool. `avg_lig_frac` is read but has no effect.

Related files

- plant.ini: plant communities for HRUs, by plant name.
- management.sch and decision tables: plant, harvest and kill operations name plants from this file.
- landuse.lum: land uses and their plant communities.
- codes.bsn: `event` switches the plant class column on; `carbon` selects whether the lignin fractions are used.

- parameters.bsn: rsd_decomp replaces a plnt_decomp of 0.
- Weather generator: monthly temperatures used for heat units to maturity.

Sources

- Reader: plant_parm_read.f90
- Type definition: type plant_db and type residue_partition_fracs in plant_data_module.f90
- Derived values and overrides: plantparm_init.f90
- Heat units to maturity: plant_init.f90
- Growth: pl_leaf_gro.f90, pl_leaf_senes.f90, pl_root_gro.f90, pl_biomass_gro.f90, pl_dormant.f90, pl_mortality.f90, pl_nfix.f90, pl_waterup.f90
- Monsoon trigger: hru_control.f90
- Residue lignin: cbn_rsd_transfer.f90, cbn_surfrsd_decomp.f90
- Calibration parameters: cal_parm_select.f90

Column	Fortran variable
name	pldb%plantnm
plnt_typ	pldb%typ
gro_trig	pldb%trig
days_mat	pldb%days_mat
bm_e	pldb%bio_e
harv_idx	pldb%hvsti
lai_pot	pldb%blai
frac_hu1, lai_max1	pldb%frgrw1, pldb%laimx1
frac_hu2, lai_max2	pldb%frgrw2, pldb%laimx2
hu_lai_decl	pldb%dlai
can_ht_max	pldb%chtmx
rt_dp_max	pldb%rdmx
tmp_opt, tmp_base	pldb%t_opt, pldb%t_base
frac_n_yld, frac_p_yld	pldb%cnyld, pldb%cpyld
frac_n_em, frac_n_50, frac_n_mat	pldb%pltnfr1, pltnfr2, pltnfr3
frac_p_em, frac_p_50, frac_p_mat	pldb%pltpfr1, pltpfr2, pltpfr3
harv_idx_ws	pldb%wsyf
usle_c_min	pldb%usle_c
stcon_max	pldb%gsi
vpd	pldb%vpdfr
frac_stcon	pldb%gmaxfr
ru_vpd	pldb%wavp
co2_hi, bm_e_hi	pldb%co2hi, pldb%bioehi
plnt_decomp	pldb%rsdco_pl
lai_min	pldb%alai_min
bm_tree_acc	pldb%laixco_tree
yrs_mat	pldb%mat_yrs
bm_tree_max	pldb%bmx_peren
ext_co	pldb%ext_coef
leaf_tov_mn, leaf_tov_mx	pldb%leaf_tov_min, pldb%leaf_tov_max
rt_st_beg, rt_st_end	pldb%rsr1, pldb%rsr2
plnt_pop1, frac_lai1	pldb%pop1, pldb%frlai1
plnt_pop2, frac_lai2	pldb%pop2, pldb%frlai2
frac_sw_gro	pldb%frsw_gro
avg_lig_frac	pldb%res_part_fracs%meta_frac
ab_lig_frac	pldb%res_part_fracs%str_frac
bg_lig_frac	pldb%res_part_fracs%lig_frac
plant class (event /= 0)	pl_class
event in codes.bsn	bsn_cc%nam1

Purpose

Fertilizer and manure applications use three databases.

- `fertilizer.frt` lists the fertilizer products: the fractions of mineral and organic nitrogen and phosphorus in each, and how much of the mineral nitrogen is ammonium. `fert` operations in `management.sch` and `fertilize` actions in decision tables name a row of this file.
- `manure_db.frt` lists the manure types. Each one points to a row of `manure_om.frt` and names optional pesticide, pathogen, metal, salt and constituent profiles. `manu` operations in `management.sch`, `manure` actions in decision tables and the manure allocation file name a row of this file.
- `manure_om.frt` gives the water, carbon and nutrient make-up of each manure.

`fertilizer.frt` is named on the `hru_parm_db` line of `file.cio` (default `fertilizer.frt`). The two manure files are not listed in `file.cio`. SWAT+ looks for them under the fixed names `manure_db.frt` and `manure_om.frt`, and leaves the manure databases empty when they are missing.

fertilizer.frt

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one product with the values below in this order. The reader does not match columns by header name. It reads the first six values and ignores the rest of the line, so the `pathogens` and `description` columns written by the SWAT+ Editor have no effect.

#	Column	Units	Default	Description
1	`name`	-	-	Name of the fertilizer, up to 16 characters. Must be unique.
2	`min_n`	kg mineral N/kg fertilizer	0.0	Fraction of the fertilizer that is mineral nitrogen (nitrate plus ammonium).
3	`min_p`	kg mineral P/kg fertilizer	0.0	Fraction of the fertilizer that is mineral phosphorus.
4	`org_n`	kg organic N/kg fertilizer	0.0	Fraction of the fertilizer that is organic nitrogen.
5	`org_p`	kg organic P/kg fertilizer	0.0	Fraction of the fertilizer that is organic phosphorus.
6	`nh3_n`	kg NH3-N/kg mineral N	0.0	Fraction of the mineral nitrogen that is ammonium. The rest is nitrate.

The fractions are by weight of the product. `min_n` + `min_p` + `org_n` + `org_p` is the nutrient content and should not exceed 1; the remainder is carrier.

For a commercial fertilizer the grade gives the nitrogen and phosphate content. The phosphate value in a grade is P2O5, so it has to be converted to P (multiply by 0.44) before it goes into `min_p`.

Fertilizer application

A fertilizer operation applies an amount in kg/ha of product. The application method row in `chem_app.ops` gives the fraction placed in the top 10 mm of soil (`surf_frac`); the rest goes into the second soil layer. In each of the two layers:

- `min_n` × (1 − `nh3_n`) of the amount is added as nitrate and `min_n` × `nh3_n` as ammonium.
- `min_p` of the amount is added as labile (soluble) phosphorus.
- With `carbon` = 0 in `codes.bsn`, half of the organic N and P goes to the fresh residue pool and half to the active humus pool.
- With `carbon` = 2, organic N and P are added to the soil organic matter pools with a carbon content of 10 times the organic N. The share that goes to the metabolic and structural litter pools depends on the C:N ratio. The split between the fresh and humus parts is `manure_c_frac` in `carbon.bsn`.

fertilizer.frt example

The first rows of `refdata/Ames_sub1/fertilizer.frt` in the SWAT+ repository:

```
fertilizer.frt: written by SWAT+ editor v2.2.2 on 2023-10-17 14:43 for SWAT+ rev.60.5.4
name                      min_n         min_p         org_n         org_p         nh3_n         pathogens
```

```
elem_n          1.00000     0.00000     0.00000     0.00000     0.00000          null
elem_p          0.00000     1.00000     0.00000     0.00000     0.00000          null
anh_nh3         0.82000     0.00000     0.00000     0.00000     1.00000          null
urea            0.46000     0.00000     0.00000     0.00000     1.00000          null
18_46_00        0.18000     0.20200     0.00000     0.00000     0.00000          null
dairy_fr        0.00700     0.00500     0.03100     0.00300     0.99000          null
```

- `urea` is 46% mineral N, all of it applied as ammonium.
- `18_46_00` is an 18-46-0 grade: 18% N and 46% P2O5, which is 20.2% P.
- `dairy_fr` is fresh dairy manure, mostly organic N.
- `elem_n` has `nh3_n` = 0, so all its nitrogen is applied as nitrate.

manure_db.frt

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one manure type with eight values.

#	Column	Description
1	`name`	Name of the manure type, up to 25 characters. Manure operations and the manure allocation file refer to it by this name.
2	`org_min`	Name of a row in `manure_om.frt` that gives the water, carbon and nutrient make-up.
3	`pests`	Name of a pesticide profile, or `null`.
4	`paths`	Name of a pathogen profile, or `null`.
5	`hmets`	Name of a heavy metal profile, or `null`.
6	`salts`	Name of a salt profile, or `null`.
7	`constit`	Name of a profile for other constituents, or `null`.
8	`description`	Free text, one word.

In 62.0.1 the `pests`, `paths`, `hmets`, `salts` and `constit` names are read but not looked up. Salt and constituent loads in manure come from the fertilizer files of the salt and constituent modules, taken from the row in the same position as the manure type. They are added only when the manure type's `org_min` names the second or a later row of `manure_om.frt`.

A manure application takes its make-up from the row of `manure_om.frt` in the same position as the manure type in `manure_db.frt` (first with first, second with second). The `org_min` name is matched against `manure_om.frt` when the file is read, but the application does not use the match. Keep the two files in the same order, with `org_min` naming the row in the same position.

manure_db.frt example

`refdata/Osu_1hru/manure_db.frt`:

```
manure_db.frt: manure database
name          org_min            pests  paths  hmets  salts  constit  description
beef_salt     mw_beef_liq        null   null   null   null   null     midwest_beef_liquid_salt
```

One manure type, `beef_salt`, using the `mw_beef_liq` make-up. `mw_beef_liq` is also the first row of `manure_om.frt` below, so the position used by the application and the `org_min` name agree. Because `org_min` names the first row, no salt or constituent loads are added when `beef_salt` is applied, whatever the salt and constituent fertilizer files contain.

manure_om.frt

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one manure make-up with nine values.

#	Column	Units	Description
1	`name`	-	Name of the make-up, up to 64 characters.
2	`frac_water`	kg water/kg manure as applied	Water fraction of the manure as applied. Must be below 1.
3	`fcbn`	kg C/kg dry manure	Carbon fraction.
4	`fminn`	kg mineral N/kg dry manure	Mineral nitrogen fraction (nitrate plus ammonium).
5	`fminp`	kg mineral P/kg dry manure	Mineral phosphorus fraction.
6	`forgn`	kg organic N/kg dry manure	Organic nitrogen fraction.
7	`forgp`	kg organic P/kg dry manure	Organic phosphorus fraction.
8	`fnh3n`	kg NH3-N/kg mineral N	Fraction of the mineral nitrogen that is ammonium.
9	`description`	-	Free text, one word.

Manure application

The amount in a manure operation is the dry mass of manure in kg/ha. The nutrient fractions apply to that dry mass. The water that comes with it, amount / (1 − `frac_water`) − amount, is added to the soil water of the two top layers. As for fertilizer, `surf_frac` of the application method in `chem_app.ops` goes to the top 10 mm and the rest to the second layer.

Mineral N and P are added the same way as for fertilizer. With `carbon` = 0, the organic N and P are split between the fresh and humus pools with a fixed fraction of 0.5. With `carbon` = 2, carbon from `fcbn` and organic N and P go to the manure and litter pools, and the split uses `manure_c_frac` in `carbon.bsn`.

manure_om.frt example

`refdata/Osu_1hru/manure_om.frt`:

```
manure_om.frt: manure organic matter database
name            frac_water   fcbn          fminn         fminp         forgn         forgp         fnh3n  c
mw_beef_liq     0.988        0.000780576   0.000311751   0.00081295    0.002110312   0.000365707   0.99   r
mw_beef_slur    0.926        0.016786571   0.002697842   0.00081295    0.002110312   0.000365707   0.99   r
```

Two make-ups of Midwest beef manure. The liquid form is 98.8% water and the slurry 92.6%. Applying 1000 kg/ha of `mw_beef_liq` adds about 82,300 kg/ha of water, or 8.2 mm.

Related files

- `management.sch`: `fert` and `manu` operations.
- Decision tables: `fertilize` and `manure` actions.
- `chem_app.ops`: application methods and the surface fraction.
- `manure_allo.mnu`: moves manure from sources to the land.
- `codes.bsn`: the `carbon` column selects how organic nutrients enter the soil.
- `carbon.bsn`: `manure_c_frac` for the dynamic carbon model.

Sources

- Readers: `fert_parm_read.f90`, `manure_db_read.f90`, `manure_orgmin_read.f90`, called from `proc_db.f90`
- Type definitions: `type fertilizer_db`, `type manure_database` and `type manure_attributes` in `fertilizer_data_module.f90`
- Name lookup: `read_mgtops.f90`
- Operations: `mgt_sched.f90`, `actions.f90`
- Application: `pl_fert.f90`, `pl_manure.f90`, `cs_fert.f90`, `salt_fert.f90`
- Fresh and humus split: `man_coef%rtof` in `carbon_module.f90`, read from `carbon.bsn` in `carbon_bsn_read.f90`

File and column	Fortran variable
fertilizer.frt `name`	`fertdb%fertnm`

File and column	Fortran variable
fertilizer.frt `min_n`, `min_p`	`fertdb%fminn`, `fertdb%fminp`
fertilizer.frt `org_n`, `org_p`	`fertdb%forgn`, `fertdb%forgp`
fertilizer.frt `nh3_n`	`fertdb%fnh3n`
manure_db.frt `org_min`	`manure_db%org_min`, index in `manure_db%iorg_min`
manure_db.frt `description`	`manure_db%descrip`
manure_om.frt columns	`manure_om%` with the same names

Purpose

`tillage.til` is the tillage database. Each row describes one tillage implement or operation: how much of the soil it mixes and how deep. A `till` operation in `management.sch`, a `till` or `tillage` action in a decision table, and a puddling operation all name a row of this file.

One row has a special role. A row named `biomix` sets the biological mixing efficiency and depth used by the dynamic carbon model (see Biological mixing).

The file is named on the `hru_parm_db` line of `file.cio` (default `tillage.til`). If the file is missing or named `null`, the database is empty and tillage operations find no row.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one operation with the values below in this order. The reader does not match columns by header name. It reads the first six values and ignores the rest of the line, so the `description` column written by the SWAT+ Editor has no effect.

#	Column	Units	Default	Typical range	Description
1	`name`	-	-	-	Name of the tillage operation, up to 16 characters. Management schedules and decision tables refer to the operation by this name, so names must be unique. The name `biomix` is reserved for biological mixing.
2	`mix_eff`	fraction	0.0	0.0 to 1.0	Mixing efficiency. The fraction of the soil, residue, nutrients and other constituents in each layer within the mixing depth that is mixed uniformly over that depth. The rest stays in its original layer.
3	`mix_dp`	mm	0.0	0 to 750	Depth of mixing caused by the operation.
4	`rough`	mm	0.0	-	Random roughness. Read but not used.
5	`ridge_ht`	mm	0.0	-	Ridge height. Read but not used.
6	`ridge_sp`	mm	0.0	-	Ridge interval or row spacing. Read but not used.

Tillage mixing

On the day of a tillage operation SWAT+ takes `mix_eff` and `mix_dp` from the named row. For each soil layer whose bottom is within `mix_dp`, the fraction `mix_eff` of the layer's mass is mixed. A layer that straddles `mix_dp` contributes in proportion to the part of it above the mixing depth. The mixed material is then spread over the mixing depth in proportion to layer thickness. Soil water, texture, mineral and organic nutrient pools, residue and constituents are all mixed this way.

A mixing depth greater than 0 and less than 10 mm is raised to 11 mm. A row with `mix_dp` of 0 mixes nothing.

The `carbon` column of `codes.bsn` decides what else happens. With `carbon` = 2 (dynamic carbon), the operation also raises the tillage factor of each mixed layer, which speeds the decomposition of soil organic matter for `till_eff_days` days. The tillage factor curve uses `tillmix_a`, `tillmix_b` and `tillmix_c` from `carbon.bsn`. With `carbon` = 0, only the mixing is done.

A puddling operation (`pudl` in `management.sch`) names a row of `puddle.ops` and a row of this file. When the HRU has more than 1 mm of ponded water, the soil is mixed with a routine for flooded soils. Otherwise the mixing is the same as for an ordinary tillage operation.

Biological mixing

Biological mixing is the slow mixing of the soil by roots, worms and other organisms. How it is set depends on the carbon model.

With `carbon` = 2 in `codes.bsn`, the `biomix` row of `tillage.til` sets it. Its `mix_eff` is the biological mixing efficiency and its `mix_dp` the maximum biological mixing depth. Biological mixing runs every day in layers above 0 °C. After a tillage operation the efficiency drops to zero and builds back up over the following days at a rate set by `bio_consol` in `carbon.bsn`. If the file has no `biomix` row, the efficiency is 0.2 and the depth 50 mm. A `biomix` row with `mix_eff` of 0 turns daily biological mixing off.

With `carbon` = 0, the `biomix` row is not used for mixing. Biological mixing then happens once a year, at the end of the year, with the efficiency in the `bio_mix` column of `hydrology.hyd` and a fixed depth of 50 mm (or the soil depth, if smaller).

Surface roughness

`rough`, `ridge_ht` and `ridge_sp` are not used. The random roughness used for surface depression storage is 20 mm for every HRU in 62.0.1, whatever tillage is done.

Example

`refdata/Osu_1hru/tillage.til` in the SWAT+ repository:

```
tillage.til: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                    mix_eff        mix_dp         rough      ridge_ht      ridge_sp  description
puddle                  0.980       150.000        00.000         0.000         0.000    rice_paddy_puddlin
```

One operation named `puddle`, used for rice paddy puddling. It mixes 98% of the soil in the top 150 mm. The file has no `biomix` row; Osu_1hru runs with `carbon` = 0 in `codes.bsn`, so biological mixing comes from `hydrology.hyd` and the missing row has no effect.

The first lines of `refdata/Ames_sub1/tillage.til`, as shipped:

name	mix_eff	mix_dp	rough	ridge_ht	ridge_sp	description
bedderd	0.55	150	35	0	0	bedder(disk)
bedders	0.55	150	30	0	0	beddershaper
beddhipr	0.65	150	35	0	0	bedderdisk-hipper
beddkrow	0.85	100	40	0	0	bedderdisk-row
bedrollr	0.25	50	5	0	0	bedroller

beetcult	0.25	25	15	0	0	beetcultivator
biomix	0.01	50	0	0	0	biological_mixing

This file has no title line. SWAT+ skips the first two lines whatever they contain, so it takes the header as the title and the **bedderd** row as the header, and **bedderd** is not in the database. A schedule that names **bedderd** would not find it. The other 98 rows are read normally; for example **mldboard** (moldboard plow) mixes 95% of the top 250 mm. Ames_sub1 runs with **carbon** = 2, so its **biomix** row sets a daily biological mixing efficiency of 0.01 over the top 50 mm.

Related files

- management.sch: **till** and **pudl** operations name a row of this file.
- Decision tables: **till** and **tillage** actions name a row of this file.
- puddle.ops: puddling parameters that go with the tillage row.
- codes.bsn: the **carbon** column selects the mixing routine and how biological mixing is set.
- carbon.bsn: tillage factor coefficients, **till_eff_days** and **bio_consol** for the dynamic carbon model.
- hydrology.hyd: the **bio_mix** column sets biological mixing when **carbon** = 0.

Sources

- Reader: till_parm_read.f90
- Type definition and biological mixing defaults: tillage_data_module.f90
- Name lookup for schedules and decision tables: read_mgtops.f90, dtbl_lum_read.f90
- Operations: mgt_sched.f90, actions.f90
- Mixing: mgt_newtillmix_cswat0.f90, mgt_newtillmix_cswat1.f90, mgt_newtillmix_wet.f90, mgt_tillfactor.f90
- Biological mixing: mgt_biomix.f90, soil_nutcarb_init.f90, hru_control.f90, time_control.f90
- Random roughness: zero0.f90, swr_depstor.f90

Column	Fortran variable
name	tilldb%tillnm
mix_eff	tilldb%effmix
mix_dp	tilldb%deptil
rough	tilldb%ranrns
ridge_ht	tilldb%ridge_ht
ridge_sp	tilldb%ridge_sp

carb_coefs.cbn is no longer read.

Removed

SWAT+ 62.0.1 ignores a copy of **carb_coefs.cbn** left in the run directory. Its settings now come from carbon.bsn and carbon_lyr.bsn.

Replacement files

The dynamic carbon model is switched on with **carbon** = 2 in codes.bsn. In earlier revisions the value was 1; in 62.0.1 the value 1 is reserved for the C-FARM model, which is not implemented. With **carbon** = 0 soil carbon is static and neither replacement file is read.

With **carbon** = 2 SWAT+ needs two files, both with a title line, a header line and data rows like other SWAT+ inputs:

- carbon.bsn: one row of 28 basin-wide values. It is named on the basin line of file.cio (default carbon.bsn).
- carbon_lyr.bsn: decomposition rates and CO2 fractions, one row for the top soil layer (**layer** 1) and one for the layers below (**layer** 2). Its name is built from the carbon.bsn name by adding **_lyr** before the extension, so it is not listed in file.cio.

If either file is missing, or carbon.bsn does not have 28 values, SWAT+ stops with an error. There is no fallback to built-in values.

An optional file carbon_layers.prt sets how many soil layers the per-layer carbon outputs include. Without it, the outputs cover the largest number of layers of any HRU.

Keyword migration

Keyword in carb_coefs.cbn	Replacement
cbn_diagnostics	Removed. Carbon outputs are switched on per file in print.prt with the hru_cb_* and lsu_cb_* rows.
hp_rate, hs_rate, microb_rate, meta_rate, str_rate, microb_top_rate, hs_hp	Columns of the same names in carbon_lyr.bsn.
a1co2, asco2, apco2, abco2	Columns of the same names in carbon_lyr.bsn.
org_frac (4 values)	init_seq, init_microb, init_slow, init_passive in carbon.bsn.
prmt_21	koc_c in carbon.bsn.
prmt_44	solc_ratio in carbon.bsn.
till_eff_days	till_eff_days in carbon.bsn.
rtof	manure_c_frac in carbon.bsn.
cbn_consolidation_factors (2 values)	bio_consol, till_consol in carbon.bsn.
cbn_factor_approaches (2 values)	tmpf_eqn, watf_eqn in carbon.bsn.
tn, top, tx	t_cbn_min, t_cbn_opt, t_cbn_max in carbon.bsn.
zz_bmix_coefs (3 values)	bmix_a, bmix_b, bmix_c in carbon.bsn.
zz_emix_coefs (3 values)	tillmix_a, tillmix_b, tillmix_c in carbon.bsn.
photo_degrade_factor	sfc_rsd_photodeg in carbon.bsn.
nmbr_soil_test_layers, soil_test	Removed, no replacement.

Six values that were fixed in the code are now columns of carbon.bsn: n_act_frac (fraction of humus nitrogen kept in the active pool), cnr_cap and cnr_ref (cap and reference C:N ratio of surface residue, formerly 500 and 25), cpr_cap and cpr_ref (cap and reference C:P ratio, formerly 5000 and 200), and mathers_method (method for the initial slow humus pool).

Example

refdata/Ames_sub1 in the SWAT+ repository has both replacement files, converted from its former carb_coefs.cbn:

```
carbon.bsn: derived from carb_coefs.cbn (basin carbon scalars)
          init_seq       init_microb         init_slow      init_passive             koc_c        solc_rat
              0.95              0.02              0.44              0.54            1000.0               0

carbon_lyr.bsn: derived from carb_coefs.cbn (per-layer carbon coefficients)
           layer         hp_rate         hs_rate     microb_rate       meta_rate        str_rate microb_to
               1         1.2e-05        2.92e-04          0.0164          0.0405          0.0107
               2         1.2e-05        1.81e-04            0.02          0.0507          0.0134
```

Related files

- carbon.bsn and carbon_lyr.bsn: the replacement inputs.
- codes.bsn: the carbon switch.
- print.prt: carbon output switches.
- Carbon outputs: the files written by the dynamic carbon model.

Sources

- Readers: carbon_bsn_read.f90 (reads carbon.bsn and carbon_lyr.bsn), carbon_layers_read.f90, both called from proc_bsn.f90

- Variables: carbon_module.f90
- Carbon switch: bsn_cc%cswat in basin_module.f90
- Output switches: basin_print_codes_read.f90
- File name slot: in_basin%carbon_bsn in input_file_module.f90

Purpose

treatment.trt is a file name kept from the SWAT2012 treatment table. Project setup tools can still write it into a project folder, but SWAT+ 62.0.1 does not open it, and file.cio has no slot for it. Its contents have no effect on a run, and the file can be left empty or deleted.

Water treatment in SWAT+ is part of water allocation. Treatment plants are defined in water_treat.wal, and the composition of treated water in om_treat.wal.

Format

None. The file is not read.

Example

Both reference datasets in the SWAT+ repository, refdata/Ames_sub1 and refdata/Osu_1hru, contain an empty treatment.trt.

Related files

- Water allocation: water_treat.wal and om_treat.wal, the treatment inputs SWAT+ reads.
- file.cio: lists the input files SWAT+ reads.

  Sources

  - File name slots read from file.cio: input_file_module.f90, readcio_read.f90
  - Treatment inputs: water_treatment_read.f90 (water_treat.wal), om_treat_read.f90 (om_treat.wal)

Purpose

manure_allo.mnu sets up manure allocation: manure sources with a monthly production, and HRUs that receive manure from them when a decision table in lum.dtl asks for it. The file name is fixed. If the file is missing, manure allocation is off.

SWAT+ 62.0.1 reads this file and checks the decision tables every day, but applies no manure to any HRU. To apply manure, use a fertilize action in lum.dtl or a manu operation in management.sch.

Format

Free format, whitespace-separated.

- Line 1: title (skipped).
- Line 2: number of allocation objects.
- Then, for each allocation object:
  1. header line (skipped);
  2. object line: name, rule_typ, src_obs, trn_obs;
  3. header line (skipped);
  4. src_obs source lines;
  5. header line (skipped);
  6. trn_obs demand lines.

Object line

#	Column	Description
1	`name`	Allocation object name.
2	`rule_typ`	Allocation rule. Read but not used.
3	`src_obs`	Number of source lines.
4	`trn_obs`	Number of demand lines.

Source lines

#	Column	Units	Description
1	`num`	-	Source number, 1 to `src_obs`. The line is stored at this position.
2	`mois_typ`	-	Moisture class, `wet` or `dry`.
3	`manure_typ`	-	Manure name in `fertilizer.frt`.
4	`lat`	degrees	Latitude of the source.
5	`long`	degrees	Longitude of the source.
6	`stor_init`	t	Manure in storage at the start of the simulation.
7	`stor_max`	t	Storage capacity.
8 to 19	`prod_jan` … `prod_dec`	t/month	Manure produced in each month, January to December.

Demand lines

#	Column	Description
1	`num`	Demand number, 1 to `trn_obs`. The line is stored at this position.
2	`ob_typ`	Receiving object type: `hru`, `muni` (treatment) or `divert` (diversion out of the basin). Only `hru` demands are linked to a decision table.
3	`ob_num`	Number of the receiving object.
4	`dtbl`	Name of the decision table in `lum.dtl` that requests manure for this HRU, or `null`.
5	`right`	Manure right, `sr` (senior) or `jr` (junior).

The decision table requests manure with a `manure_demand` action: `obj_num` is the source number, `option` the application method in `chem_app.ops`, `const` the amount in t/ha and `const2` the maximum number of requests per year (see `*.dtl`).

Example

One allocation object with one source and one HRU demand:

```
manure_allo.mnu: example
1
name            rule_typ    src_obs  trn_obs
farm_a          fixed             1        1
num  mois_typ  manure_typ  lat       long      stor_init  stor_max   prod_jan  feb  mar  apr  may  jun  ju
  1  wet       beef_fr     41.99   -93.62          0.0       500.0    5.0  5.0  6.0  6.0  7.0  7.0  7.0  7
num  ob_typ    ob_num  dtbl                right
  1  hru            1  apply_manure         sr
```

With a table `apply_manure` in `lum.dtl` whose first action is `manure_demand` with source 1, method `broadcast` and 2 t/ha, the intent is to apply 2 t/ha of `beef_fr` from the source whenever the table's conditions are met. In 62.0.1 the request is recorded but nothing is applied.

Related files

- fertilizer.frt: the manure named in manure_typ.
- *.dtl: the decision tables named on demand lines and the manure_demand action.
- chem_app.ops: application methods.

Sources

- Reader: manure_allocation_read.f90, called from main.f90.in after dtbl_lum_read
- Types: manure_allocation_module.f90 (manure_allocation, manure_source_objects, manure_demand_objects)
- Daily calculation: mallo_control.f90, called from time_control.f90
- manure_demand action: actions.f90, cross-walk in dtbl_lum_read.f90
- HRU decision table slot: hru_dtbl_actions_init.f90
- Output: manure_source_output.f90, manure_demand_output.f90, called from command.f90

Column	Fortran variable
object line	mallo%name, %rule_typ, %src_obs, %trn_obs
source manure_typ	mallo%src%manure_typ (index in mallo%src%fertdb)
source prod_jan … prod_dec	mallo%src%prod_mon(1:12)
demand dtbl	mallo%trn%dtbl (index in mallo%trn%dtbl_num and hru%man_trn_dtbl)
manure_demand obj_num, const, option	manure_amt%src_obj, %app_t_ha, %app_method

The management schedules are in management.sch. Four small inputs referred to from other pages follow it: puddle.ops, transplant.ops, co2_yr.dat and object.prt.

management.sch

management.sch contains the management schedules. Each schedule has a list of operations on fixed dates or heat-unit fractions, a list of automatic operations (decision tables in lum.dtl), or both. An HRU uses the schedule named in the mgt column of its land use in landuse.lum. The file name is the second entry on the lum line of file.cio (default management.sch).

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Then, for each schedule:

1. The schedule line: name, numb_ops, numb_auto.
2. numb_auto lines, each naming one decision table in lum.dtl.
3. numb_ops operation lines.

Column	Description
name	Schedule name, referenced by mgt in landuse.lum.
numb_ops	Number of scheduled operation lines.
numb_auto	Number of automatic operation lines.

Automatic operations

Each automatic operation line names a decision table in lum.dtl. Every day, SWAT+ checks each listed table for every HRU that uses the schedule. A name that is not in lum.dtl is not reported.

Three generic tables can be shared by several schedules with different crops. For these, the crop name follows the table name on the same line, and the plant, harvest and harvest_kill actions of the table use it instead of their option column:

Table name	Line content
pl_hv_summer1	table name and one crop
pl_hv_winter1	table name and one crop
pl_hv_summer2	table name and two crops; the crop used is the one for the current rotation year

Operation lines

#	Column	Units	Description
1	op_typ	-	Operation code (see below).
2	mon	-	Month of the operation, or 0.
3	day	-	Day of the month, or 0.
4	hu_sch	fraction	Heat-unit fraction that triggers the operation, or 0.
5	op_data1	-	First name (see below).
6	op_data2	-	Second name (see below).
7	op_data3	varies	Number: amount, days or override (see below).

op_typ	Operation	op_data1	op_data2	op_data3
plnt	plant	plant name in the HRU's plant community	name in transplant.ops	> 0 to start from the transplant entry
harv	harvest	plant name, or all	name in harv.ops	residue harvest efficiency (0 uses harv_eff)
hvkl	harvest and kill	plant name, or all	name in harv.ops	residue harvest efficiency
kill	kill	plant name, or all	-	-
till	tillage	name in tillage.til	-	-
fert	fertilizer	name in fertilizer.frt	name in chem_app.ops	amount, kg/ha
manu	manure	name in manure_db.frt	name in chem_app.ops	amount, kg/ha
pest	pesticide	pesticide name listed in constituents.cs	name in chem_app.ops	amount, kg/ha (multiplied by app_eff)
graz	grazing	name in graze.ops	-	number of grazing days
burn	burning	name in fire.ops	-	-
swep	street sweeping	name in sweep.ops	-	-
irrm	irrigation	name in irr.ops	-	-
irpm	irrigation of a rice paddy	name in irr.ops	-	-
irrp	start, adjust or stop paddy irrigation	name in irr.ops	water source: cha, res, aqu or unlim	target ponding depth, mm (see below)
pudl	puddling	name in tillage.til	name in puddle.ops	-
weir	set paddy weir height	weir name in weir.res	-	weir height, mm
dwm	drainage water management	-	-	tile drain depth, mm
cnup	change curve number	absval, abschg, pctchg or relchg	-	value
mons	monsoon season	-	-	0 or 1
skip	end of a rotation year	-	-	-

Notes on individual operations:

- plnt, harv, hvkl and kill act on the plant in the HRU's plant community (plant.ini) whose name equals op_data1. A plant that is already growing is not planted again; the event is logged as PLANT_ALREADY_GROWING.
- fert on an HRU with standing water puts the fertilizer into the water.
- pest looks the name up in the pesticide list of constituents.cs when the operation runs, so the pesticide must be simulated as a constituent.
- irrm and irpm apply the depth in irr.ops (see irr.ops).
- irrp turns on automatic irrigation of a paddy: water is added whenever the ponding depth falls below 90% of the target. A positive op_data3 is the target depth; a negative value uses amt_mm of the irr.ops entry; zero

turns paddy irrigation off. The irr.ops entry also gives the efficiency, runoff fraction and salt and nitrate concentrations of the water.

- cnup changes the curve number for moisture condition II: set it to the value (absval), add the value (abschg), change it by the value in percent (pctchg), or multiply it by 1 + value (relchg). The result is kept between 35 and 95.
- mons applies to plants with a moisture_gro trigger in plants.plt. With op_data3 1 it marks the start of the monsoon season. With 0 it restarts the phenology of such plants that are not growing.
- Codes not in this table (for example pcom or prtp) are read but do nothing.

Timing

Operations run in the order of the lines, one rotation year after another. A skip line ends a rotation year. After the last line the schedule starts again at the first line, so a schedule with n skip lines is an n-year rotation.

An operation runs on the day its mon and day match the calendar date, or, when hu_sch is greater than 0, on the first day the heat-unit fraction exceeds hu_sch. The heat-unit fraction is that of the plant named in op_data1 while it grows, and the base-zero fraction of the year otherwise. Give each operation either a date or a heat-unit fraction and set the other to 0. Within a rotation year, lines must be in chronological order: SWAT+ waits for the date of the current line before it moves on.

When the schedule reaches a skip line it waits there until the end of the year: 31 December for HRUs in the northern hemisphere, day 181 for HRUs in the southern hemisphere. At the start of the run, SWAT+ begins at the rotation year given by the plant community in plant.ini and skips the operations dated before the first simulated day.

The operations performed are written to mgt_out.txt when mgtout is y in print.prt.

Example

refdata/Ames_sub1/management.sch in the SWAT+ repository, first lines:

```
management.sch file AMES
                          NAME NUMB_OPS NUMB_AUTO OP_TYP        MON        DAY HU_SCH   OP_DATA1    OP_DAT
mgt_01                             305         1
                                                irr_year_irr
                                                    till          4          1    0.0    chisplow        n
                                                    plnt          4         28    0.0       corn         n
                                                    irrm          5         15    0.0       sprinkler_ilm
                                                    fert          6          1    0.0       null       ele
                                                    hvkl         10          7    0.0       corn        gr
                                                    skip          0          0      0       null         n
```

Schedule mgt_01 has 305 operation lines and one automatic operation, the irr_year_irr table of lum.dtl (see the *.dtl example). In the first rotation year it tills with chisplow on 1 April, plants corn on 28 April, irrigates once on 15 May, and harvests and kills the corn with the grain harvest operation on 7 October.

For a fert line, 62.0.1 reads the fertilizer name from op_data1 and the application method from op_data2. The fert line above has null in op_data1 and the fertilizer name elem_n in op_data2. No fertilizer is named null, so the line matches no entry in fertilizer.frt and applies nothing; no warning is written. Every fert line in this file has the same layout, so as written the Ames schedules apply no fertilizer in 62.0.1. A line that applies 181.4 kg/ha of elem_n in 62.0.1 reads fert 6 1 0.0 elem_n broadcast 181.4, with broadcast an entry in chem_app.ops.

Two schedules from refdata/Osu_1hru/management.sch, one using a scheduled manu operation together with decision tables and one using only decision tables:

```
rice140_rot                      1          7
     plow
     weir60r
     weir0
     pad_irr_r
     puddle
```

```
     pl_hv_rice
     fert_paddy
manu      5          1      0     beef_salt   broadcast  12.9927
agrl_rot                        0          2
    pl_hv_agro
    fert_stress
```

As shipped, the `manu` line finds `beef_salt` in manure_db.frt and `broadcast` in chem_app.ops, so 12.99 kg/ha of manure is applied on 1 May. The schedule has one operation line and no `skip`, so the line runs every year.

Related files

- landuse.lum: selects a schedule by name in its `mgt` column.
- *.ops, tillage.til, fertilizer.frt: databases the operation lines name.
- *.dtl: the decision tables listed as automatic operations.
- plant.ini: plant community and initial rotation year.

puddle.ops

Puddling parameters for rice paddies. A puddling operation (`pudl` in `management.sch`, or the `puddle` action in lum.dtl) mixes the soil with a tillage operation, lowers the hydraulic conductivity of the paddy's upper soil, and sets the sediment concentration of the ponded water. The file name is fixed.

Free format. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one entry, read by position.

#	Column	Units	Description
1	`name`	-	Puddling operation name.
2	`hydcon_mm/h`	mm/h	Hydraulic conductivity of the upper soil after puddling. 0 leaves it unchanged.
3	`sed_ppm`	ppm	Sediment concentration of the ponded water after puddling.
4	`orgn_ppm`	ppm	Organic N concentration after puddling. Read but not used.
5	`sedp_ppm`	ppm	Organic P concentration after puddling. Read but not used.
6	`no3_ppm`	ppm	NO3-N concentration after puddling. Read but not used.
7	`solp_ppm`	ppm	Soluble P concentration after puddling. Read but not used.
8	`nh3_ppm`	ppm	NH3 concentration after puddling. Read but not used.
9	`no2_ppm`	ppm	NO2 concentration after puddling. Read but not used.

The tillage operation is named in `op_data1` of the `pudl` line (or the `name` column of the `puddle` action). If the paddy has more than 1 mm of water, the mixing is done in the ponded water; otherwise it is ordinary tillage.

`refdata/Osu_1hru/puddle.ops` as shipped:

```
name                hydcon_mm/h   sed_ppm      orgn_ppm      sedp_ppm     no3_ppm       solp_ppm       nh
high_eff            0.00000         10000       0.00000     800.00000     10.00000       20.00000        1
med_eff             0.00000         10000       0.00000     500.00000     10.00000       20.00000        1
low_eff             0.0         10000       0.00000     100.00000     10.00000       20.00000        1.000
```

The file has no title line. SWAT+ skips the column header as the title and the `high_eff` line as the header, so 62.0.1 reads only `med_eff` and `low_eff`. An operation that names `high_eff` finds no entry and uses an entry with all values zero: conductivity unchanged and no sediment added. The Osu dataset's `puddle` decision table names `med_eff`, so its puddling runs with a sediment concentration of 10000 ppm and unchanged conductivity (`hydcon_mm/h` is 0).

transplant.ops

Transplant database. A planting can start a plant from a transplanted state instead of from seed: a `plnt` line in `management.sch` with a `transplant.ops` name in `op_data2` and `op_data3` greater than 0, or a `plant` action in `lum.dtl` with the name in its `fp` column. The file name is fixed. Up to revision 61 it was `transplant.plt`; 62.0.1 reads only `transplant.ops`, so a project with a `transplant.plt` file needs it renamed.

Free format. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one entry, read by position.

#	Column	Units	Description
1	`name`	-	Transplant name.
2	`lai_ini`	m^2/m^2	Leaf area index at transplanting. Read but not used.
3	`bm_ini`	kg/ha	Plant biomass at transplanting.
4	`phu_acc_ini`	fraction	Fraction of potential heat units already accumulated.
5	`fr_yrmat`	fraction	Fraction of the years to maturity already reached (perennials).
6	`pop`	plants/m^2	Plant population. Sets the maximum leaf area index through the population curve in `plants.plt`. 0 uses the maximum LAI in `plants.plt`.

`refdata/Osu_1hru` contains the file under its old name:

```
transplant.plt
      NAME       LAI_INI     BM_INI    PHU_ACC_INI    FR_YRMAT      POP
   tr_rice140         0.1        0.2      0.2             1            100
```

62.0.1 does not open `transplant.plt`, so in this dataset the transplant database is empty. The `plant` actions in its `lum.dtl` that name `tr_rice140` find no entry, no message is written, and the rice is planted from seed with the initial values of an ordinary planting.

co2_yr.dat and co2.out

Atmospheric CO2 concentration by year. `co2_yr.dat` is optional; its name is fixed. SWAT+ builds one CO2 value per simulation year and writes the series to `co2.out`.

`co2_yr.dat` layout:

Line	Content
1	title (skipped)
2	number of year records n
3	header (skipped)
4 to $n + 3$	calendar year and CO2 concentration (ppm), one year per line, consecutive years in increasing order

How the series is built:

- If `co2_yr.dat` is missing or empty, every year uses `co2` from `parameters.bsn` (default 400 ppm).
- If the last record's year is at or before the first simulation year, every year uses the last record's value.
- Otherwise the records are aligned to the simulation by the first record's year. Years before the first record use the first value, and years after the last record repeat the last value.

The alignment counts records from the first one, so the years must be consecutive.

`co2.out` has the header `YR  CO2(ppm)` and one line per simulation year with the calendar year and the CO2 value used.

object.prt

Hydrograph and state output for individual objects. Each line asks for one variable of one object, written daily to a separate file. The file name is the third entry on the `simulation` line of `file.cio` (default `object.prt`; `null` for none).

Free format. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one request.

#	Column	Description
1	`id`	Sequence number, 1, 2, 3, … in order.
2	`obj_typ`	Object type: `hru`, `hlt` (HRU-lte), `ru` (routing unit), `res`, `cha` (channel), `sdc` (SWAT-DEG channel), `exc` (export coefficient), `dr` (delivery ratio) or `out` (outlet).
3	`obj_typ_no`	Object number within its type. For `hru` with `sol_water` or `solnut_ly`, 0 writes every HRU.
4	`hyd_typ`	Variable (see below).
5	`filename`	Output file name, up to 26 characters.

`hyd_typ`	Output
`tot`	total outflow hydrograph
`rhg`	recharge (percolation) hydrograph
`sur`	surface runoff hydrograph
`lat`	lateral flow hydrograph
`til`	tile flow hydrograph
`sol_water`	soil water of each layer
`solnut_ly`	soil N and P pools of each layer (no header line)
`solnut_pr`	soil N and P pools of the profile (no header line)
`plant`	plant status
`cha_fp`	channel and floodplain water balance

Each requested file gets a header line (except `solnut_ly` and `solnut_pr`) and one line per day with day of year, month, day, year, object type and name, followed by the values.

Sources

- `management.sch` readers: `mgt_read_mgtops.f90` (schedule and automatic operation lines), `read_mgtops.f90` (operation lines and name lookups)
- Schedule types: `mgt_operations_module.f90`
- Running scheduled operations: `mgt_operatn.f90`, `mgt_sched.f90`, called from `hru_control.f90`
- Starting operation and year-end `skip` handling: `plant_init.f90`, `time_control.f90`
- Automatic operation lookup: `proc_cond.f90`; generic crop tables: `actions.f90`
- Curve number change: `chg_par.f90`
- `puddle.ops`: `mgt_read_puddle.f90`
- `transplant.ops`: `plant_transplant_read.f90`, `mgt_transplant.f90`, type `plant_transplant_db` in `plant_data_module.f90`
- CO2: `co2_read.f90`, `bsn_prm%co2` in `basin_module.f90`
- `object.prt`: `object_read_output.f90`, `obj_output.f90`, type `object_output` in `hydrograph_module.f90`
- File name slots: `in_lum%management_sch` and `in_sim%object_prt` in `input_file_module.f90`

File	Column	Fortran variable
`management.sch`	`numb_ops`, `numb_auto`	`sched%num_ops`, `sched%num_autos`
`management.sch`	automatic operation line	`sched%auto_name`, `sched%auto_crop`
`management.sch`	`op_typ`, `mon`, `day`	`mgt_ops%op`, `%mon`, `%day`
`management.sch`	`hu_sch`	`mgt_ops%husc`
`management.sch`	`op_data1`, `op_data2`, `op_data3`	`mgt_ops%op_char`, `%op_plant`, `%op3`
`puddle.ops`	`hydcon_mm/h`	`pudl_db%wet_hc`
`transplant.ops`	`lai_ini`, `bm_ini`, `phu_acc_ini`	`transpl%lai`, `%bioms`, `%phuacc`

File	Column	Fortran variable
co2_yr.dat	year, CO2	co2_inc%co2_yr%iyr, %co2
object.prt	obj_typ, obj_typ_no, hyd_typ	ob_out%obtyp, %obtypno, %hydtyp